

Web Accessibility for GTM Dashboards

October 2024 | Phoenix Baldino-Clarke

Why are we here?



Based on feedback collected from our internal dashboard users, our dashboards need some tweaks that improve accessibility.

Feedback

Our internal customers include people with disabilities on the web who are struggling with some of our design choices

Colors and shapes are not as clear to others with visual impairments and screen readers when navigating our dashboards

These are universal issues that appear across multiple GTM dashboards, even if not specified by the users



Opportunity

Learn more about web accessibility and how the tools you build can affect user experience

Apply deliberate design strategies that promote overall usability of our Tableau dashboards

Take advantage of internal and external tools that help us meet digital accessibility standards

Feedback from Our Dashboard Users: Usability Challenges

01

Actual Impact Exists

There are more color blind users within our internal customers than you may realize. They are struggling to interpret the data we have visualized in our dashboards.

02

Color Blindness

Certain colors are difficult to see when mixed. Yellow and light green combinations are particularly challenging for our users.

03

Stoplight Challenge

The red, green, amber color combination is commonly used but difficult to distinguish for people with color blindness. The importance of the colors are lost for visually impaired users.

04

Low Contrast

Different color contrasts are needed when the text on the page is too close in color to the background.

05

Less Tooltip Importance

There is too much critical information in tooltips, which can only be accessed when a user hovers in specific locations – instead of being available on-screen. Tooltips require fine motor control.

06

Shapes & Sizes

Using the same shape but different colors makes the meaning too difficult to distinguish for users with visual impairments or screen readers.



28.7%

Per the CDC, more than 1 in 4 adults in the US have some type of disability

Disability Categories



Visual

Blindness, low vision, color-blindness.

***Our focus today**



Auditory

Deafness and hard-of-hearing.



Motor

Inability to use a mouse, slow response time, limited fine motor control.



Cognitive

Learning disabilities, distractibility, inability to remember or focus on large amounts of information.

What is color blindness?



Color blindness is a color vision deficiency (CVD) that makes it hard for people to distinguish between different colors. It can affect the ability to distinguish between shades of red and green, or even to see all colors across the spectrum.

People with color blindness usually adapt to living with it and don't have problems with everyday activities.

99%

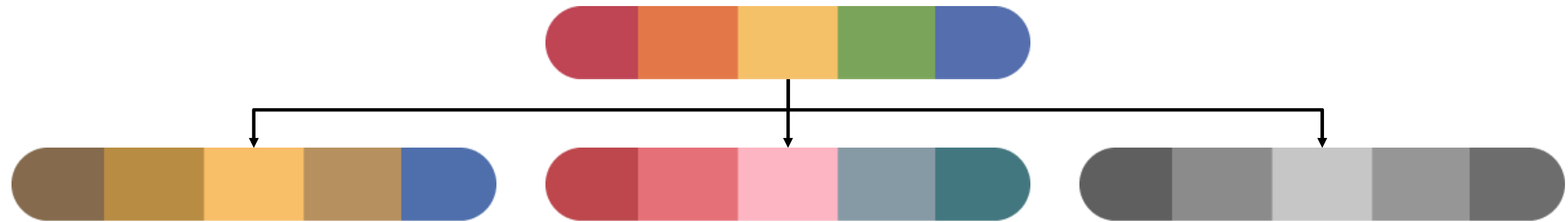
The most common type of color blindness is red-green.
99% of all color-blind people suffer from red-green color blindness.

1ⁱⁿ12

1-in-12 men and 1-in-200 women have color vision deficiency.

Types of Color Blindness

There are **seven official diagnosis types** of color blindness: four in the red-green category, two in blue-yellow, and one describing the vision completely lacking in color.



Red-Green color blindness

- Protanopia (aka red-blind) and Deuteranopia (green-blind): individuals can't distinguish between red and green
- Protanomaly (aka red-weak): individuals can see some shades of red
- Deuteranomaly (aka green-weak): individuals can see some shades of green

Blue-Yellow color blindness

- Tritanopia (aka blue-blind): individuals can't distinguish between blue and green, purple and red, yellow and pink
- Tritanomaly (aka blue-weak): individuals can see some shades of blue and yellow

Complete color blindness

- Individuals with this condition see no color at all — their world exists in black and white
- Occurs in one out of 33,000 people

Other Relevant Web Accessibility Challenges and User Frustrations

JPMC Digital Accessibility Knowledge Base (DAKB) Personas

01

Space Between Items

I get confused when **cluttered environments** and/or flashing text make it difficult to stay focused on my tasks.

02

Small Links / Hovers

I have difficulty reading small text, blocks of text, and clicking on **small links (target areas)** because I have tremors in my hand.

03

Low Contrast

I struggle with light gray text color and white backgrounds because it often just looks like a blank white space because the **contrast is so low**.

04

Overlapping Content

All content I view online must be **significantly enlarged** which can cause challenges when the proximity of elements is too far apart when enlarged, cut off, or overlaps.

05

Horizontal Scrolling

I become fatigued when I must work harder to read **content that requires me to scroll horizontally** (left-to-right).

06

Screen-Reader Friendly

Arthritis in my hand causes pain when using a standard mouse. I am trying to learn to use **voice commands and my smart speaker** so it takes more time to complete tasks.

07

Visual & Tab Order

I am annoyed when an interface doesn't **follow a logical pattern**. I'm not stupid, but learning new patterns can be a challenge.

08

Color Blindness

As someone with Protanopia (can't perceive red light), I have trouble when **red is used as the only error indicator** to let me know my password is incorrect when I log in.

What can we do to be more inclusive in our designs?

The Firm



Dashboard Developers



Standards

JPMC follows the W3C's globally recognized standards to provide content that's compatible with the most popular assistive technologies and web browsers. <https://www.chase.com/digital/resources/accessibility>



Policies

Accessibility requirements for accessible **employee-facing digital content** are documented in the firm-wide [Disability Inclusion Standards](#). Requirements for accessible customer-facing digital content are documented in the firm-wide [ADA Policy](#) and [ADA Standards](#).



Training

The [Digital Accessibility Team provides](#) **firm-wide educational resources**, topic-based and role-based instructor-led/self-paced eLearning trainings for various roles ([UX](#), [Development](#), & [Testing](#)), and a newsletter.

Awareness

The foundation of any commitment to accessibility is awareness of the barriers. Few developers oppose the broad concept of inclusive design; many simply do not **perceive the barriers faced by users with disabilities**.



Leadership

If we demonstrate leadership through a commitment to web accessibility, chances are high that the our web content will be accessible. Developers will make content accessible if they feel that **the effort is expected, recognized and rewarded**.



Intentionality

The idea of accessibility is backed by the firm's policies and procedures. Let's ensure it becomes part of the daily routine of designing dashboards. Now that we **know better**, let's **do better**.



We have knowledge and awareness. Where do we start?

Dashboard Developers

Focus on Top 10 most-used dashboards.

Use TabAuto ([go/tabauto](#) > Analysis) to identify the most-used dashboards that you own. These are most likely to have an impact on users with visual disabilities. Focus efforts on dashboards affecting key business objectives.

Phoenix will provide TabAuto usage analysis results after this training.



Review top dashboards for specified pain points.

Make a list of changes that will improve usability and functionality of your dashboard with accessibility in mind. Use the empathy, knowledge, and awareness gained to generate a to-do list that addresses concerns of users.

Phoenix can perform dashboard review and provide recommendations upon request.



Use color blindness simulator for a different lens.

Use a free, external color blindness simulator or an internal tool provided for developers to examine your dashboards through a different lens. Note color choices that are meant to convey meaning but not showing up well.

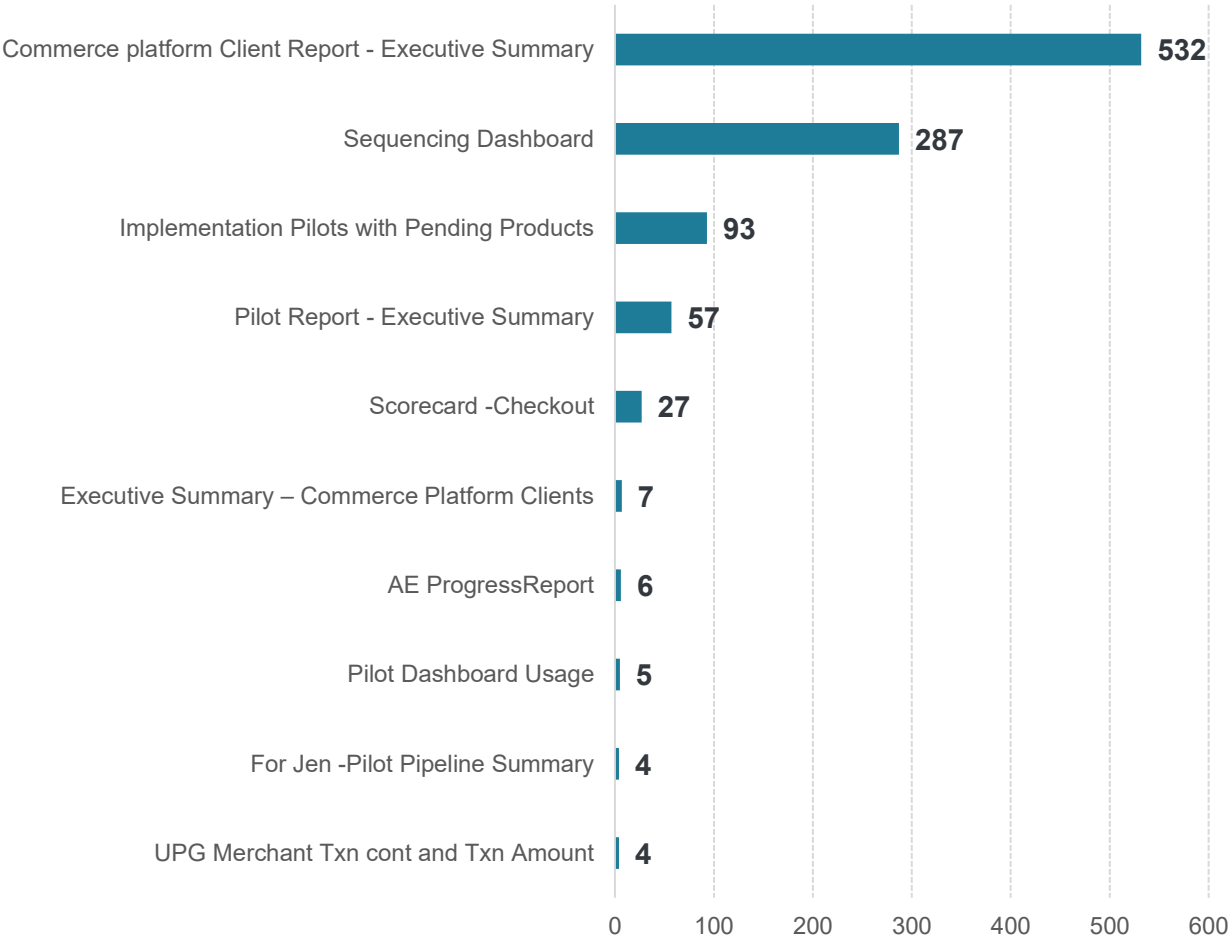


Make changes that improve digital accessibility.

Notify your primary business stakeholder(s) of the changes you plan to apply. Remember that some changes may require use notification or education to ensure data is still interpreted as intended.



TOP 10 MOST-USED TABLEAU WORKBOOKS* IN MCEA PROJECT
AUG 2024 | with Developer Usage Removed



Source: TabAuto Usage Summary

** A single Tableau workbook can contain 1 or more dashboards in the same file.*

Recommended Changes for Accessibility

Turning Feedback into Progress

Regardless of whether you are developing a dashboard for internal or external users, applying established design techniques and best practices will ensure your end users receive a high-quality design result that provides the highest impact to drive decision and innovation.



- 1 Use JPMorgan brand colors when assigning to shapes and objects in Tableau dashboards.**
Your Tableau installation already includes multiple color palettes specific to *JPMC Brand Colors*.



- 2 Follow the [Tableau COE Style Guide](#) for typography best practices.**
This style guide includes best practices and recommendations for use of color (backgrounds, highlight vs no highlight, base chart color, etc), typography guide (font sizes and styles), and so on.



- 3 Use Icons and Graphics from the brand assets catalog instead of default shapes.**
Icons can be downloaded directly from the brand library in various file types compatible with Tableau Desktop.



- 4 Reduce visual noise to prevent cognitive overload.**
Unnecessary borders, gridlines within charts, and strategic use of labels helps users assert focus and interpret data more quickly.



- 5 Avoid low contrast text: grey text on light backgrounds**
When using text, use only black text on white backgrounds or white text on black backgrounds to create maximum contrast.

Recommended Changes for Accessibility (cont.)



6 Use Color sparingly and align color usage with need to assert eye focus

Instead of making every mark of a chart a different color to differentiate and emphasize volume, consider what happens if those colors look the same to the end user. Does the mark with the highest volume have other indicators that highlight volume? Are end users able to quickly spot what you are trying to emphasize with color?



7 For stoplight colors, include black text with colors applied to mark or shape – not colored text

Since Red, Yellow, and Blue fall into multiple combinations of color blindness types, using those colors in a single chart introduces more opportunity for the meaning to be lost visually. Be sure to add dark text to describe the color, such as black or navy.



8 Remove command buttons from tooltips

End users with difficulty in their motor skills (e.g. severe arthritis), may not click as quickly or precisely as expected. As a result, including command buttons with the selection of a mark means someone could accidentally filter and NOT know how to recover from it.

Tips & Tricks: Tableau and PowerPoint Demos

Click below to open demo workbook in Tableau Desktop.

Open Tips-and-Tricks_Accessibility_Workbook.twbx

[GTM SharePoint Drive](#) > Documents > Phoenix Clarke > **Accessibility Tips & Tricks** > Tips-and-Tricks_Accessibility_Workbook.twbx

Resources for Developers

JPMorganChase

Digital Accessibility Knowledge Base (DAKB)

Internal resource contains videos that reveal the user experience of each [Accessibility Personas](#) and a handy [Accessibility Design Checklist](#) to ensure your design is accessible.



JPMorganChase Brand Toolbox (<https://jpmcbrandtoolbox.com/>)

Website that contains the brand guidelines, logos, and color schemes for applying the **JPMorganChase** brand to your design.

You already have an account if login credential were emailed to you in SEP.



Chase Brand Toolbox (<https://www.chasebrandtoolbox.com/>)

Website that contains the **Chase** brand guidelines, logos, and color schemes at Toolbox > Design menu > Color.

You must request access through the link on site login page.



Color Blindness Simulators

- [Pilestone](#) - Allows you to upload an image instead of requiring URL to dashboard
- [Colblinder](#) - allows for image upload and provides options for different types of color blindness
- [Venngage](#) - not appropriate for internal usage but has good standard examples



Contrast Checker

- [WebAIM Contrast Checker](#)



Recommended Articles

- [Colors to Avoid for Color Blind Users: An Intro for Designers](#)
- [Tableau Guide: Build Accessible Dashboards](#)
- [Integrating Interface Design Principles in Data Visualizations](#)